Enhancing Student Support through Speech Emotion Recognition and Dialogue Summarization

Rohit Ray

Abstract

Human–computer interaction increasingly depends on systems that can recognise and respond to human affect. This research report presents an end-to-end speech emotion recognition (SER) pipeline purpose-built for naturalistic, one-to-one student–teacher interactions acquired from public YouTube recordings. The pipeline integrates audio pre-processing (voice activity detection, denoising, loudness normalisation, and segmentation), deep models for emotion classification (Wav2Vec2-base, Wav2Vec2-XLS-R, HuBERT-base, and a lightweight CNN), automatic speech recognition (Whisper), and abstractive dialogue summarisation to produce concise, actionable insights. To mitigate class imbalance, we employ weighted sampling and class-weighted cross-entropy; model selection is driven by accuracy and macro-F1, complemented by confidence calibration. Across comparative experiments, HuBERT-base achieved the most balanced validation performance and the sharpest confidence calibration, making it the preferred model for deployment. The resulting system outputs emotion labels with calibrated confidence scores, transcripts, and short summaries, enabling efficient review of moments of heightened affect or potential distress without replaying entire sessions. We discuss ethical considerations, limitations, and opportunities for real-world adoption in student wellbeing monitoring.

Keywords

Speech Emotion Recognition; HuBERT; Wav2Vec2; Whisper; Dialogue Summarisation; Affective Computing; Education; Student Wellbeing; Confidence Calibration.

1.INTRODUCTION

1.1 Introduction Chapter

The emotional state of students plays a key role in their academic performance, engagement, and overall well-being. For international students, adapting to new cultural, academic, and social settings can lead to increased stress, anxiety, and feelings of isolation. Educational institutions are acknowledging the importance of monitoring emotional well-being to provide timely support. However, traditional observation methods often depend on subjective measures, are time-consuming, and are not practical on a large scale.

With the rapid growth of artificial intelligence (AI), speech-based emotion recognition has emerged as a promising way to assess emotional states without being intrusive. Spoken language shows emotional cues through tone, pitch, rhythm, and intensity. This enables computational models to infer emotional states without relying solely on facial expressions or written text.

This dissertation examines the use of deep learning methods to detect emotions in one-on-one student–teacher interactions by analyzing audio collected from online videos. The study builds a dataset from YouTube recordings of educational interactions and evaluates four models: three transformer-based architectures (Wav2Vec2, Wav2Vec2-LG, and HuBERT) and a custom convolutional neural network (CNN) baseline. Comparative experiments reveal that the HuBERT model is the most effective, and it is later applied in a prototype inference pipeline for real-time deployment.The research addresses two primary objectives:

1. To create a strong, complete system for collecting, processing, and analysing audio data to identify emotional states confidently.
2. To investigate how this system can be a scalable tool for early detection of emotional distress, allowing schools to support students' mental health proactively.

By offering real-time, objective emotion analysis, this work adds to the growing area of affective computing in education and addresses an urgent need for technology-driven monitoring of student well-being.

1.2 Research Aims and Objectives

The main goal of this research is to create and assess a system that can identify emotional states in recorded student-teacher interactions through audio analysis.

The specific objectives include:

1. Acquire recordings from YouTube and convert them into mono 16 kHz WAV audio files.
2. Apply denoising and silence filtering to improve signal quality and remove segments with little information.
3. Segment audio into fixed-length clips of 20 seconds to allow for consistent emotion analysis.
4. Generate weak labels for training data using a pre-trained speech emotion recognition model with confidence thresholds.
5. Fine-tune multiple deep learning models, including Wav2Vec2, Wav2Vec2-LG, and HuBERT, along with a CNN baseline trained on mel-spectrograms.
6. Evaluate and compare the models based on prediction accuracy and confidence scores using unseen data to find the best-performing model.
7. Create an interactive inference interface where users can upload YouTube or Google Drive video links. The system will extract

audio and produce emotion predictions with confidence scores.

8. Add features for transcription and summarization, storing summaries in text format for further analysis.

1.3 Research Beneficiaries

The findings of this dissertation hold potential value for several groups:

- Universities and higher education institutions, especially those supporting international students and prioritizing student well-being.
- Mental health professionals and student counselors, who can utilize emotion insights for early intervention.
- Educational policymakers, who may apply the outcomes to design strategies that enhance institutional mental health support.

1.4 Project Constraints and Ethical Considerations

Project Constraints

The project was implemented using Python and Google Colab, both of which are freely available. Model training, inference, and evaluation were conducted using open-source libraries within Google Colab's free or Pro tiers, requiring no direct budget allocation.

Key limitations include:

1. Variability in audio quality, background noise, and microphone conditions, which may affect detection accuracy.
2. Cross-cultural and linguistic differences in emotional expression, which may reduce generalizability.
3. The computational demands of transformer models, making large-scale or real-time deployment difficult without powerful hardware.
4. Model evaluation focused only on English-language student-teacher interactions, which may not translate easily to other languages or contexts.

Ethical Considerations

1. The dataset was collected solely from publicly available YouTube videos of educational interactions, avoiding private or sensitive recordings.
2. No personally identifiable information was gathered, so formal ethical approval was not necessary.
3. All sources were credited properly, following academic fair-use guidelines

2. Related Work

2.1 Introduction

Speech Emotion Recognition (SER) has become a key research area in artificial intelligence (AI). It brings together psychology, linguistics, and machine learning to identify human emotions

through speech. Unlike text-based sentiment analysis, which only looks at word usage, SER uses prosodic and acoustic features like pitch, tone, rhythm, and intensity (El Ayadi et al., 2011). These features often provide clearer indications of emotional states and highlight subtleties that text alone may miss.

In education, SER is especially useful. It gives insights into student engagement, motivation, and well-being. Emotional states impact learning outcomes. Positive emotions can lead to greater persistence and creativity, while negative emotions, like frustration or anxiety, can disrupt focus and hinder knowledge retention (Pekrun, 2014). This is particularly true for international students, who deal with extra cultural and social challenges (Sawir et al., 2012).

This chapter reviews previous work on SER. It outlines the development of methods, datasets, tools, and applications. It also points out challenges and research gaps, highlighting the contribution of this dissertation.

2.2 Concept of Emotion Detection in Speech

Emotions are essential to human communication. They affect not only what we say but also how we say it and how it is understood. While text conveys clear meaning, vocal qualities provide hidden emotional signals. Factors like pitch, loudness, speed, and pauses are strong indicators of how someone feels. For example, anger often shows up as louder speech and a higher pitch, whereas sadness typically appears through slower speech and less energy (Schuller et al., 2018).

Detecting emotions in speech involves looking at these vocal and acoustic features. This helps classify speech into specific categories like happy, sad, angry, or neutral, or into continuous emotional scales such as arousal and valence. Both methods are common in research, each with unique benefits. The categorical method offers clear labels for classification, while the dimensional method captures emotional nuances better (Russell, 1980).

In educational contexts, speech-based emotion recognition (SER) has many benefits. Unlike facial recognition, which needs high-quality video and can invade privacy, SER can be done discreetly using audio from natural interactions like classroom discussions or online meetings (Cummins et al., 2018). This makes it ideal for large-scale, real-time monitoring of student well-being in digital learning settings.

2.3 Evolution of Speech Emotion Recognition Approaches

2.3.1 Early Acoustic Feature-Based Methods

The earliest SER research focused on handcrafted acoustic features from signal processing techniques. Commonly extracted features included:

- Mel-Frequency Cepstral Coefficients (MFCCs): capturing the short-term power spectrum of sound.
- Prosodic Features: pitch, energy, duration, and rhythm.
- Voice Quality Features: jitter, shimmer, formants, and harmonic-to-noise ratio.

These features were typically modeled using statistical methods like Gaussian Mixture Models (GMMs) and Hidden Markov Models (HMMs) (Ververidis and Kotropoulos, 2006). While they worked well in controlled lab settings, these methods struggled in noisy, real-world recordings. Variability in microphones, speakers, and environments caused distortions. Their dependence on manual feature engineering also limited their ability to scale and adapt to different datasets.

2.3.2 Traditional Machine Learning Approaches

With the rise of machine learning in the 2000s, researchers used classifiers like Support Vector Machines (SVMs), Random Forests (RF), k-Nearest Neighbors (k-NN), and Naïve Bayes. These methods provided better generalization than traditional statistical models and were efficient for medium-sized datasets (Schuller et al., 2009). However, they still relied heavily on manually designed features and had a hard time capturing long-range temporal dynamics in conversational speech. For example, while SVMs performed well on standard datasets, their effectiveness dropped in spontaneous, noisy, or multilingual situations.

2.3.3 Deep Learning Approaches

The arrival of deep learning changed the landscape of SER research. Convolutional Neural Networks (CNNs) allowed for the direct analysis of spectrograms by treating them as two-dimensional images. This effectively captured local time and frequency patterns related to emotional expression (Trigeorgis et al., 2016). Recurrent Neural Networks (RNNs) and Long Short-Term Memory networks (LSTMs) further improved recognition by modeling sequential dependencies in speech. This made them well-suited for dialogues and longer recordings (Lee and Tashev, 2015).

Despite these advancements, CNNs and RNNs still relied on spectrogram representations, which needed prior feature engineering. Additionally, their performance often suffered due to small dataset sizes and sensitivity to noise. This limited their use in large-scale, real-world applications.

2.3.4 Transformer-Based Models

The introduction of transformer-based architectures has changed speech emotion recognition (SER). Models like Wav2Vec2 (Baevski et al., 2020), XLS-R (Conneau et al., 2021), and HuBERT (Hsu et al., 2021) use self-supervised

pretraining on large, unlabelled speech datasets. They learn contextualized representations straight from raw waveforms, removing the need for manually crafted acoustic features. Fine-tuning these models on SER datasets has consistently produced top-notch results, even in noisy settings.

Transformers also work well for multilingual and cross-domain applications. For instance, XLS-R has been pretrained on 128 languages, allowing effective cross-lingual transfer in emotion recognition. Likewise, HuBERT has shown strength in extracting speech units that are helpful not just for recognition but also for affective computing. This shows a shift towards general-purpose pretrained speech models that can be easily adjusted for various downstream tasks, including SER.

2.3.5 Multimodal and Transfer Learning

Another emerging trend is the integration of multimodal data sources. Combining audio with text through automatic speech recognition or video through facial expressions improves performance by capturing multiple channels of emotional expression (Poria et al., 2017). Transfer learning allows SER systems to use large pretrained models from other speech or language tasks. This has further improved generalization across languages, speakers, and recording conditions (Pepino et al., 2021).

However, multimodal systems also bring challenges. They require more computing power and raise additional privacy concerns, especially in educational and healthcare settings. Despite these limitations, multimodal fusion is an important research direction for developing more complete and accurate emotion recognition systems.

2.4 Datasets for Speech Emotion Recognition

Datasets are central to SER research, as they shape both the development of models and the evaluation of their performance. Benchmark datasets have enabled standardized comparisons between methods, but their characteristics also impose limitations on generalizability.

2.4.1 Benchmark Datasets

Several widely used datasets form the foundation of SER research:

- **RAVDESS (Ryerson Audio-Visual Database of Emotional Speech and Song):** Contains acted emotional expressions across speech and song modalities, recorded by professional actors (Livingstone and Russo, 2018).
- **CREMA-D:** A crowd-acted dataset where multiple annotators labeled speech samples for emotional categories, improving inter-rater reliability (Cao et al., 2014).
- **IEMOCAP (Interactive Emotional Dyadic Motion**

Capture): One of the most extensively used multimodal corpora, featuring scripted and improvised dyadic conversations annotated with categorical and dimensional emotions (Busso et al., 2008).

- **EMO-DB:** Among the earliest emotional speech datasets, containing German acted speech samples that have served as a benchmark for early acoustic feature-based methods (Burkhardt et al., 2005).

These datasets have played an essential role in advancing SER methodologies, providing consistent platforms for testing feature extraction techniques, machine learning algorithms, and deep learning models.

2.4.2 Limitations of Acted Datasets

Despite their importance, most benchmark datasets rely on acted emotions, which differ greatly from natural emotional expressions. Acted emotions are often exaggerated and lack the subtle nuances found in real-world interactions (Latif et al., 2020). This limits the real-world applicability of models trained only on this data, as they may focus too much on "idealized" emotions instead of the complex emotional signals we see in everyday speech.

Another major issue is the size of the datasets. Many SER corpora contain only a few thousand samples, restricting the training of data-hungry deep learning and transformer-based models (Sahu et al., 2021). Small datasets also worsen class imbalance problems, where certain emotions like "neutral" or "happy" dominate, while less common feelings like "fear" or "disgust" are underrepresented.

2.4.3 Naturalistic Datasets

To address these shortcomings, researchers have increasingly turned to natural data sources such as:

- Call center recordings, where speech shows stress, frustration, or satisfaction in real customer interactions.
- Clinical interviews, which capture emotional states connected to psychological conditions like depression or anxiety (Cummins et al., 2018).
- Online platforms like YouTube, which provide a lot of varied and spontaneous audio data.

YouTube, in particular, has become a valuable source because of its size and variety. Unlike laboratory datasets, YouTube recordings reflect different demographics, accents, and environments. However, they also bring challenges, including background noise, inconsistent audio quality, and overlapping speech. Preprocessing, like denoising, silence removal, and segmentation, is critical for turning raw YouTube audio into usable SER training datasets.

While these datasets are more challenging to process, they offer greater ecological validity and better represent real-world conditions in education, healthcare, and customer service. By training models on these varied, noisy datasets, researchers hope to improve robustness and generalization beyond controlled lab settings.

2.5 Tools and Frameworks

Modern Speech Emotion Recognition (SER) research has been greatly supported by the rapid development of open-source tools and frameworks. These resources simplify tasks such as data preprocessing, feature extraction, model development, and evaluation, making cutting-edge methods more accessible to researchers and practitioners.

2.5.1 Feature Extraction Tools

A key step in SER is extracting meaningful acoustic and prosodic features from raw audio. Several tools are widely used:

- OpenSMILE (Munich Open-Source Multimedia Feature Extractor): It provides a complete set of predefined feature sets, such as INTERSPEECH challenges feature configurations, making it a standard for benchmarking. It supports extracting prosodic, spectral, and voice-quality features, which are essential in both traditional and deep learning methods (Eyben et al., 2010).
- Praat: This phonetic analysis tool is frequently used in linguistics. Praat allows precise analysis of pitch, formants, intensity, and temporal features. Its scripting capabilities enable automation across large datasets.
- Librosa: This Python library is designed for audio analysis. Librosa is particularly popular in machine learning pipelines because it can generate Mel-spectrograms, chroma features, and tempo information (McFee et al., 2015).

Together, these tools support both low-level feature extraction, like MFCCs, energy, and zero-crossing rate, and higher-level descriptors, like prosody and rhythm. These features form the foundation of SER pipelines.

2.5.2 Deep Learning Frameworks

The rise of deep learning has led to several popular frameworks, including:

- PyTorch: Known for its dynamic computation graph and flexibility, PyTorch is widely used

for research-oriented SER. It is particularly good for rapid prototyping and model experimentation.

- TensorFlow/Keras: This framework provides strong deployment options and is popular in production environments. It is also used in academic SER studies. Its connection with TensorBoard enables detailed monitoring of model performance.

These frameworks offer optimized versions of neural network layers, loss functions, and GPU acceleration. This helps in training complex models like CNNs, RNNs, and transformers for SER tasks.

2.5.3 Pretrained Models and Transfer Learning Platforms

The availability of pretrained speech models has sped up SER research. The Hugging Face Transformers library has become a key resource, offering implementations of Wav2Vec2, HuBERT, and XLS-R. Researchers can fine-tune these models on SER datasets with little computational cost compared to training from scratch (Wolf et al., 2020). These pretrained models provide general speech representations, allowing effective transfer learning across different domains and languages. Their inclusion in open-source libraries ensures that even researchers with limited resources can access top models.

2.5.4 Data Processing and Evaluation Tools

Efficient preprocessing and evaluation are crucial in SER:

- FFmpeg: A versatile multimedia framework that supports tasks like resampling, channel conversion (e.g., stereo to mono), denoising, and segmentation. It is commonly used to standardize audio inputs before feature extraction or model training.
- Scikit-learn: Offers strong tools for preprocessing (e.g., normalization, dimensionality reduction) and evaluation metrics (accuracy, F1-score, confusion matrices). It is often used in deep learning workflows for consistent model evaluation.

2.5.5 Democratization of SER Research

The growing availability of open-source frameworks has made SER research more accessible. In the past, researchers depended on proprietary tools or limited academic licenses, which restricted experimentation. Now, libraries like PyTorch, Librosa, and Hugging Face enable reproducibility, transparency, and scalability. This openness encourages

collaboration and drives progress in affective computing, as models and datasets can be shared and validated between institutions.

2.6 Challenges in Speech Emotion Recognition

Despite notable progress in recent years, Speech Emotion Recognition remains a challenging research area. Advances have come from better models and larger datasets, yet several key obstacles still exist. These challenges limit the reliability, scalability, and practical use of SER systems.

2.6.1 Data Imbalance

Most SER datasets are imbalanced. Certain emotions, like happiness or neutrality, dominate, while rarer feelings such as fear or disgust are underrepresented. For instance, in the widely used IEMOCAP dataset, there are significantly more samples of neutral and anger compared to sadness or disgust (Busso et al., 2008). This imbalance results in biased classifiers. Models may show high overall accuracy but perform poorly on the less frequent emotions.

To address this, researchers have looked into methods like data augmentation (e.g., pitch shifting, time-stretching, noise injection), generating synthetic data using Generative Adversarial Networks (GANs), and using class-weighted loss functions (Latif et al., 2020). However, augmentation techniques can distort emotional signals, and GAN-generated speech often lacks the authenticity of natural emotional variations.

2.6.2 Noise and Variability

Real-world audio data frequently contains background noise, overlapping conversation, echoes, and differences in microphone quality. These factors obscure emotional cues and lower recognition accuracy. For example, call center or online meeting data often has overlapping speech and inconsistent volume levels, making segmentation and labeling more complex.

While denoising algorithms and speech enhancement techniques can help, they may also suppress emotional signals found in pitch and intensity. Self-supervised models like Wav2Vec2 and HuBERT have shown promise in dealing with noisy inputs because of their strong pretraining on large, diverse datasets (Hsu et al., 2021). Still, achieving consistent performance in unpredictable acoustic conditions remains a challenge.

2.6.3 Cultural and Linguistic Variability

Emotional expression is heavily shaped by cultural and linguistic norms. For instance, the same prosodic cue (like rising pitch) can mean politeness in one language but anger in another (Pell et al., 2009). Likewise, high-context cultures may express emotions more subtly compared to low-context cultures, making universal classification difficult.

Models trained on single-language datasets (like English) often struggle to

generalize to different languages, highlighting the need for multilingual datasets and cross-cultural transfer learning. Pretrained models like XLS-R, which support over 100 languages, show promise in tackling this issue, but evaluations across diverse cultural groups are still limited.

2.6.4 Subjective Labeling

Labeling emotions is inherently subjective. Even human annotators often disagree, especially on unclear categories like neutral, mixed emotions, or low-intensity affects. Inter-annotator agreement rates for SER datasets are frequently below 70%, raising concerns about the reliability of the labeled data (Devillers et al., 2005).

To manage this subjectivity, researchers have proposed probabilistic labeling (assigning distributions instead of fixed labels), crowdsourced annotation to capture varied perspectives, and continuous emotion models (like valence-arousal space) that better reflect subtle emotional states. However, these approaches complicate model training and evaluation since many machine learning algorithms are designed for categorical classification.

2.6.5 Computational Demands

Transformer-based models like Wav2Vec2 and HuBERT offer top-notch accuracy but are resource-intensive. They require GPUs with significant memory and lengthy training times. This limits their use in real-time settings such as classrooms or mental health monitoring. For example, fine-tuning Wav2Vec2 on modest hardware can take several hours per training cycle, making large experiments impractical without cloud computing resources.

Researchers are looking into model compression techniques (like knowledge distillation, pruning, and quantization) along with lighter architectures such as CNN-based models to boost efficiency. Nonetheless, there is a trade-off between model complexity, accuracy, and real-time operation that remains unresolved.

2.6.6 Generalizability and Real-World Deployment

Another challenge is ensuring that SER systems can generalize well. Models trained on controlled datasets typically struggle to work effectively in real-world environments with natural, spontaneous speech. This raises concerns about ecological validity, particularly in education and healthcare, where incorrect classifications could lead to serious issues.

To ensure robustness, systematic cross-dataset evaluations, domain adaptation methods, and real-world pilot studies are necessary. However, such validation requires considerable resources and is often overlooked in academic research focused on achieving high benchmark scores.

2.6.7 Summary of Challenges

Overall, SER research faces interconnected challenges related to data

issues, labeling, cultural differences, noise, and computational costs. Tackling these challenges will need a mix of innovative methods (like self-supervised learning, multimodal fusion, and probabilistic labeling) and careful assessment of model generalization. Progress in these areas will shape whether SER evolves from a promising field into reliable, scalable systems used in education, healthcare, and more.

2.7 Applications of Speech Emotion Recognition

Speech Emotion Recognition has many applications in various fields, showing its adaptability and importance in society. As models have become more accurate and reliable, SER has been incorporated into research and real-world systems.

2.7.1 Education

In education, SER is being tested to track student engagement and well-being. For instance, recognizing stress, confusion, or disengagement in students during online classes can help teachers adjust their methods in real-time. Automated systems could identify students who are emotionally distressed, allowing for early interventions by school counsellors or academic advisors (D'Mello and Graesser, 2015). With the growth of digital classrooms and blended learning, SER offers scalable strategies for improving student support without intrusive methods like constant video recording.

2.7.2 Healthcare

Healthcare is a highly significant application area. Small changes in vocal tone and prosody are linked to conditions like depression, anxiety, and neurodegenerative diseases such as Parkinson's and Alzheimer's (Low et al., 2020). SER systems can aid clinicians by providing early-warning signals based on patient speech during routine visits or telemedicine calls. For mental health care, ongoing monitoring through SER could promote preventive care by spotting deterioration before crises happen. However, using patient data raises important ethical issues related to consent, confidentiality, and bias in diagnoses.

2.7.3 Customer Service

In customer service, SER improves call center analysis and chatbot communication by recognizing customer feelings in real-time. For example, noticing frustration or anger during a call can lead to transferring the call to a human operator, enhancing satisfaction and loyalty (McTear et al., 2016). Companies also use SER to review customer-agent interactions for training and quality assurance. Although this benefits business efficiency, concerns about employee monitoring and emotional surveillance emphasize the need for clear policies and ethical measures.

2.7.4 Human–Computer Interaction (HCI)

Human–Computer Interaction is increasingly integrating SER to make systems friendlier and more responsive. Virtual assistants like Alexa, Siri, and Google Assistant could change their replies based on user emotions, leading to more natural and supportive interactions. In gaming and entertainment, SER can adjust experiences to match player moods, boosting engagement. Research in this field aligns with affective computing, aiming to connect human communication with machine understanding.

2.7.5 Security and Forensics

In high-stress situations such as emergency services, aviation, or law enforcement, SER can examine vocal cues for signs of panic, aggression, or deceit. For instance, analyzing emergency calls can help dispatchers gauge urgency and allocate resources effectively (Jiang et al., 2019). In forensic settings, emotion recognition can aid in profiling suspects or assessing witness credibility. However, implementing SER in security raises significant ethical and legal issues, particularly regarding privacy, potential misuse, and the risk of false positives.

2.7.6 Ethical Considerations in Applications

While SER's versatility highlights its potential impact, challenges specific to each domain must be carefully managed. In education and healthcare, privacy and consent are crucial. In business and security settings, fairness, transparency, and accountability are equally important. These factors stress the need for strong ethical frameworks alongside technological progress.

2.8 Research Gap

Despite significant advancements, several gaps remain in the SER literature, creating a need for further research.

- Overreliance on Acted Datasets: Most studies depend on acted datasets like RAVDESS and EMO-DB. While beneficial for controlled experiments, they lack ecological validity, which limits how well the models apply to real-world speech.
- Limited Comparative Analyses: Few studies systematically compare transformer-based models like HuBERT and Wav2Vec2 with lighter CNNs or hybrid models in the same experimental setup. This complicates assessing the trade-offs between accuracy and efficiency.
- Insufficient End-to-End Pipelines: Existing research often separates preprocessing, model training, and inference. Fully integrated systems capable of handling noisy raw data, running inference, and producing clear outputs have not been explored enough.

- Underexplored Educational Applications: While SER has been studied thoroughly in healthcare and customer service, its use in monitoring student well-being is still developing. Few studies have looked at large-scale, real-world datasets from educational settings, particularly in multicultural or multilingual contexts.

This dissertation addresses these gaps by:

- Developing an automated pipeline that processes real-world YouTube recordings of student-teacher interactions.
- Comparing multiple deep learning models, including CNNs and transformer-based architectures, within the same framework.
- Launching a practical inference system that combines emotion detection with transcription and summarization, connecting academic research with real-world applications.

3.METHODOLOGY

This project uses a Waterfall project management approach. It progresses from gathering requirements to evaluating the system. The methodology combines research-oriented and engineering-oriented views to illustrate how the system was designed, built, and tested.

3.1 Requirements and Analysis

Data Needs

The main requirement was to gather and process a large number of real-world audio samples from one-on-one student-teacher meetings to train and evaluate emotion recognition models. The dataset needed to:

- Include varied speech patterns, intonations, and emotions related to educational interactions and daily life interactions.
- Be publicly available to avoid privacy or ethical issues.
- Contain English-language content to ensure consistency with model training.

The targeted emotion categories were those typically used in speech emotion recognition (SER) models, such as:

- Neutral (neu)
- Angry (ang)
- Happy (hap)
- Sad (sad)

Model Selection Rationale

Four models were chosen for evaluation:

- facebook/wav2vec2-base – This serves as a baseline transformer model for speech feature extraction.
- ehcalabres/wav2vec2-lg-xlsr-en-speech-emotion-recognition – A larger version of Wav2Vec2 fine-tuned for SER, which may improve accuracy with a bigger pretraining dataset.

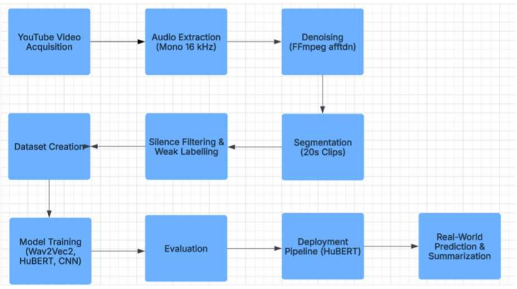
- superb/hubert-base-superb-er – This uses the HuBERT architecture and provides a different self-supervised learning method for comparison with Wav2Vec2.
- Custom CNN – This was designed for mel-spectrogram inputs and serves as a lightweight baseline for deployment in resource-limited situations.

After training and evaluation, the HuBERT model (finetuned) achieved the most balanced performance and was chosen as the final model for real-world testing.

3.2 Design

The system was designed as a multi-stage pipeline, including data acquisition, preprocessing, model training, evaluation, and deployment. This modular design was chosen over an end-to-end deep learning approach because it allows validation at every stage, ensuring that errors in preprocessing do not go undetected into later model training (cf. Schuller et al., 2018).

Workflow Overview:



A 20-second segmentation window was selected after testing shorter (5–10s) and longer (30–60s) intervals. Very short clips often lacked sufficient context for emotion classification (El Ayadi et al., 2011), while long clips tended to average out distinct emotions (Satt et al., 2017). Thus, 20 seconds provided a balance between contextual richness and emotional consistency.

The choice to use YouTube videos ensured ecological validity, reflecting real-world noise, accents, and conversational dynamics (Livingstone & Russo, 2018). However, this also introduced challenges like variable recording quality and limited metadata, concerns raised in recent SER literature on real-world datasets (Latif et al., 2020)

3.3 Data Collection

Source

YouTube videos of student-teacher interactions were selected as the primary dataset. Using publicly available recordings ensured no private or sensitive material was included, meeting ethical considerations (Kaur et al., 2022).

Acquisition Process

- Videos were downloaded using yt_dlp in Google Colab.
- ThreadPoolExecutor enabled parallel downloads, balancing efficiency with IP rate-limit constraints.

Critical Reflection

While YouTube offered naturalistic, diverse data, several limitations were identified:

1. Language bias – Videos were restricted to English, limiting cross-linguistic generalizability (Schuller & Batliner, 2014).
2. Audio quality variability – Differences in microphones and environments introduced noise that could obscure subtle emotional cues (Eyben et al., 2016).
3. Lack of metadata – Unlike curated datasets such as IEMOCAP or RAVDESS, YouTube videos provide no structured annotation, requiring weak labeling methods (Livingstone & Russo, 2018).

After preprocessing and filtering, the final dataset contained 665 clips. While this is relatively small compared to large-scale SER datasets, using pretrained transformer models reduced the need for massive labelled datasets, making this size sufficient for fine-tuning (Hsu et al., 2021).

3.4 Data Preprocessing

1. **Audio Extraction**
   Audio was extracted using FFmpeg and converted into mono-channel, 16 kHz WAV files, in line with best practices in SER preprocessing (Eyben et al., 2016).
2. **Segmentation**
   Clips were divided into 20-second segments to ensure consistency for training and prevent long recordings from diluting emotional signals (El Ayadi et al., 2011).
3. **Noise Reduction**
   The afftdn filter in FFmpeg was applied for mild denoising. This choice balanced noise reduction with preserving emotional speech cues, such as low-frequency tremors linked to sadness (Schuller et al., 2018). Excessive denoising risked losing these subtle indicators.
4. **Silence and Low-Information Filtering**
   Silence and near-silent segments were removed using waveform statistics (NumPy + Soundfile). While this improved efficiency, it raised a limitation: pauses and hesitations can themselves be emotional indicators (Grimm et al., 2007).
5. **Weak Labelling**
   A Hugging Face audio classification pipeline was used to generate initial labels. Only segments with confidence scores above 0.5 were retained, reducing noise but also discarding possibly useful ambiguous samples. This reliance on a pretrained model risked label bias, an issue raised in semi-supervised SER studies (Latif et al., 2020).

6. **Dataset Split**
   The final dataset was divided into 80% training and 20% testing subsets using a fixed random seed for reproducibility. This approach aligns with standard SER practices (Satt et al., 2017), even though it limits the chance of a dedicated validation set.

4.5 Implementation

Model Architectures

1. **Wav2Vec2 Models (facebook/wav2vec2-base, ehcalabres/wav2vec2-lg-xlsr-en-speech-emotion-recognition)**
   - Transformer-based, self-supervised models pretrained on large audio corpora(Baevski et al., 2020).
   - Fine-tuned with weighted cross-entropy to counteract class imbalance.
2. **HuBERT Model (superb/hubert-base-superb-er)**
   - Learns **hidden speech units** during pretraining, enabling robust performance in noisy conversational data(Hsu et al., 2021).
   - Showed greater adaptability to **emotion-rich spontaneous speech** compared to Wav2Vec2.
3. **Custom CNN**
   - Operated on **mel-spectrograms**, offering a lightweight alternative suitable for deployment in **resource-constrained environments**.
   - Although less accurate, it provided a **baseline for scalability and hardware efficiency(Zhang et al., 2021).**

Training Configuration

- **Batch Size:** 1–2 (limited by GPU memory).
- **Epochs:** 3 (trade-off between convergence and runtime).
- **Optimizer:** AdamW.
- **Loss Function:** Weighted cross-entropy.
- **Evaluation Strategy:** Best-model checkpointing.
- **Hardware:** Google Colab GPU.

Critical Reflection

The use of Google Colab brought both benefits (free GPU access and ease of sharing) and issues (limited session timeouts and constrained GPU memory). This required small batch sizes, which might have slowed convergence. However, the pretrained nature of transformer models helped mitigate this issue, allowing effective fine-tuning even under hardware limits (Baevski et al., 2020).

4.5.1 Transformer Architectures Used

In this research, transformer-based speech models were adopted due to their ability to capture long-range dependencies in audio and their state-of-the-art performance in speech-related tasks. The transformer architecture is fundamentally different from convolutional models, as it relies on a **self-attention mechanism** that allows the model to weigh the importance of each element in the sequence relative to others. This is particularly important in emotion recognition, where subtle changes in tone, pitch, or rhythm over time can signal different emotional states.

Wav2Vec2.0

Wav2Vec2.0 combines a convolutional feature encoder with a transformer encoder. Pretraining is performed via **masked prediction**, where parts of the latent speech representations are masked, and the model predicts them using surrounding context. This allows the model to learn robust general-purpose audio representations without requiring large amounts of labeled data. In this study, the **base model** was fine-tuned for emotion classification on the student–teacher dataset.

Wav2Vec2-XLS-R

The XLS-R model is an extension of Wav2Vec2, pretrained on **436,000 hours of speech data across 128 languages**. Its architecture mirrors Wav2Vec2 but benefits from multilingual pretraining, which enhances cross-lingual generalization. However, in this project, its performance was limited due to the small dataset size.

HuBERT (Hidden Unit BERT)

HuBERT follows the transformer encoder design but introduces a **two-stage pretraining strategy**:

1. Raw speech is clustered into discrete units (pseudo-labels).
2. The model learns to predict these masked units from context, similar to masked language modeling in NLP.
   This approach allows HuBERT to capture both acoustic and phonetic structures more effectively. In this study, HuBERT achieved the best results, demonstrating robustness to noisy classroom audio.

Comparison with CNN Baseline

A lightweight CNN model was implemented as a baseline. CNNs extract local spectral features but lack the global sequence modeling capabilities of transformers. As expected, the CNN performed worse, reinforcing the effectiveness of self-attention mechanisms in modeling emotional cues in speech.

Model	Feature Encoder	Transformer Layers	Hidden Size	FFN Dim	Attention Heads	Pretraining
Wav2Vec2-base	7 Conv (1D, strided)	12	768	3072	12	Masked latent prediction
XLS-R (300M)	7 Conv (1D, strided)	24	1024	4096	16	Multilingual masked prediction
HuBERT-base	7 Conv (1D, strided)	12	768	3072	12	Masked prediction of clustered units
CNN Baseline	4 Conv1D + 3 Pooling	–	–	–	–	Supervised only

Table 1: Model Information

3.6 Testing – Verification & Validation

Verification

Unit tests ensured correct audio preprocessing, verifying:

- Sampling rate = 16 kHz.
- Mono conversion.
- Segment length = 20 seconds.

Denoising and silence filtering were visually verified using waveform plots.

Validation

Models were evaluated using:

- **Accuracy** (overall correctness).
- **Macro F1-score** (balanced measure across classes).
- **Precision & Recall** (class-specific insights).
- **Confusion Matrix** (highlighted frequent misclassifications).

Critical Reflection

Accuracy alone was found to be misleading due to **imbalanced data** (neutral and happy more common than angry or sad). Macro F1-score provided a **fairer assessment** (Satt et al., 2017). Confusion matrices revealed that high-energy emotions like anger were sometimes misclassified as happiness, a known issue in acoustic emotion classification (Schuller & Batliner, 2014).

3.7 Deployment Pipeline

After evaluation, **the HuBERT-based model** was selected as the final deployment model due to its **balanced performance across precision, recall, and F1-score**.

Pipeline Features

1. **YouTube/Google Drive Integration** – Accepts video links, downloads, and resamples audio.
2. **Emotion Prediction** – Uses HuBERT checkpoint to generate emotion predictions with softmax confidence scores.
3. **Confidence Visualization** – Displays probability distributions across classes in a bar chart.
4. **Transcript & Summarization** – Whisper (small) generates

transcripts; DistilBART summarizes them into concise text files.

Critical Reflection

While HuBERT offered higher accuracy, it was computationally heavier than the CNN baseline, making real-time deployment challenging on low-resource devices. Whisper's transcription quality was acceptable but struggled with **accents and background noise**, limiting reliability in diverse classrooms **(Radford et al., 2022)**. Furthermore, although YouTube-based data ensured ethical compliance in research, **deploying such a system in real schools would require explicit consent and strict data privacy safeguards (Kaur et al., 2022)**.

4. Results

4.1 Overview

This chapter reports and interprets results from a speech emotion recognition (SER) pipeline evaluated on real-world student–teacher meetings sourced from YouTube. In contrast to acted corpora (e.g., RAVDESS, EMO-DB), the dataset reflects noisy, naturalistic conditions typical of deployment. Four models were assessed: (1) *facebook/wav2vec2-base*; (2) *ehcalabres/wav2vec2-lg-xlsr-en-speech-emotion-recognition* (XLS-R); (3) *superb/hubert-base-superb-er* (HuBERT); and (4) a custom spectrogram-based CNN.

4.2 Dataset and Preprocessing

Audio was extracted from 28 YouTube videos, converted to mono 16 kHz WAV, segmented into 20 s clips, denoised, and filtered for silence. Weak labels were generated with a Hugging Face classifier and retained when confidence > 0.5. The final corpus comprised 665 usable clips, split 80/20 into 532 training and 133 test items. Emotions included Neutral, Happy, Angry, and Sad, with Neutral and Sad predominating and Angry/Happy underrepresented. To mitigate the resulting imbalance, weighted cross-entropy was applied. The dataset offers higher ecological validity than acted sets (e.g., IEMOCAP, CREMA-D), but increases classification difficulty due to noise, background chatter, and varied microphones.

4.3 Model Performance

- **Wav2Vec2-base:** Validation accuracy stabilized at 64–67% with gradual loss reduction. The model frequently confused Neutral with other emotions, suggesting limited capture of subtle prosodic cues in noisy settings.

- **XLS-R (wav2vec2-lg-xlsr-en):** Achieved ~56% validation accuracy while training accuracy rose to 79%, indicating overfitting. The large multilingual pretraining and limited English-only data likely led to representation mismatch.

- **HuBERT (superb/hubert-base-superb-er):** Best overall performance with 70.7% validation accuracy and closer alignment between training and validation, indicating improved generalization on noisy, accented speech.
- **Custom CNN:** After four epochs, validation accuracy reached ~53%. While efficient for low-resource contexts, the CNN exhibited poorer calibration and struggled to model contextual dependencies beyond handcrafted mel-spectrogram features.

4.4 Confusion Matrices

Across models, **Sad** was the most reliably detected class, whereas **Neutral** was most confusable with **Happy** or **Angry**.

- **Wav2Vec2-base/XLS-R:** Pronounced confusion of Neutral with Sad/Happy.
- **HuBERT:** Stronger separation between Angry and Happy; Neutral remained ambiguous.
- **CNN:** Bias toward Sad with frequent misclassification of Happy and Neutral.
  These findings align with prior SER literature indicating low-intensity states (e.g., Neutral) share overlapping prosody with mild emotions, complicating discrimination.

4.5 Confidence Calibration

Confidence

- **Wav2Vec2-base:** Correct predictions accumulate at high confidence (>0.75); mid-range (0.4–0.6) unreliable.
- **XLS-R:** Weak calibration; incorrect predictions often appear at high confidence.
- **HuBERT:** Sharp calibration; most correct predictions occur at very high confidence (0.9–1.0).
- **CNN:** Substantial overlap in the 0.4–0.6 band, indicating weak decision boundaries.
  For educational deployment, calibration is critical to user trust; HuBERT's reliability favors it despite only moderate accuracy gains.

4.6 Comparative Discussion and Implications

Wav2Vec2-base provided stable but limited nuance capture; XLS-R overfit the small domain-specific set; the CNN offered efficiency with notable accuracy/calibration trade-offs. **HuBERT emerged as the most suitable for real-world classroom audio**, balancing accuracy, robustness to noise, and confidence reliability. The study underscores the value of ecologically valid data and the necessity of class imbalance handling. Future work should expand balanced, in-the-wild datasets; explore domain-adaptive pretraining; and incorporate calibration-

aware training (e.g., temperature scaling) to further enhance trustworthiness in educational applications.

4.7 Classification Report Analysis

Per-class precision/recall/F1 clarify trade-offs across four emotions (angry, happy, neutral, sad). *wav2vec2-base* attains 66.9% accuracy and weighted F1=0.67, excelling on angry (F1=0.83) and sad (0.71) but underperforming on happy (0.37) and moderate on neutral (0.56), suggesting difficulty with subtle cues. *XLS-R* yields 56.4% accuracy, F1=0.56, with high recall for happy/sad (0.77/0.82) but very low precision (happy=0.19), leading to many false positives; angry (F1=0.32) and neutral (0.47) are weak. *HuBERT* performs best—70.7% accuracy, weighted F1≈0.70—with strong angry (0.87), solid sad (0.73), improved happy (0.60), and moderate neutral (0.57); macro/weighted averages indicate balanced behavior. The *CNN* baseline is weakest (53.4% accuracy, weighted F1≈0.53): good on sad (0.773) but poor on happy (0.286) and angry (0.33). Overall, high-arousal emotions (angry/sad) are more reliably detected than subtle/overlapping states (happy/neutral).

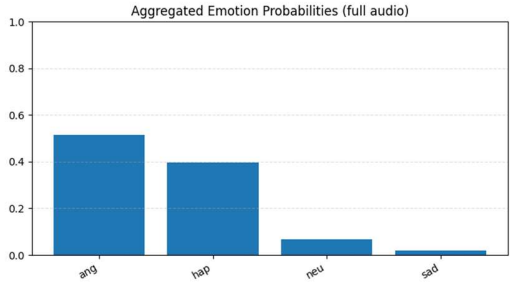
4.8 Discussion of Model Performance

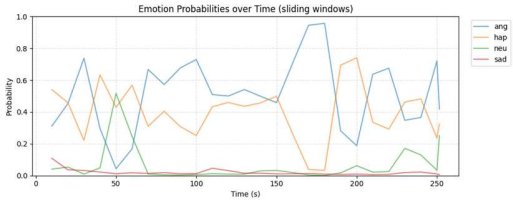
Wav2Vec2-base shows stable training with limited representational depth, hampered by noisy conversational data and neutral–happy ambiguity. *XLS-R* overfits: training accuracy rises to 79% while validation stagnates near 56%, reflecting capacity–data mismatch and domain shift (multilingual pretraining vs. small, English, classroom domain). *HuBERT*'s hidden-unit discovery pretraining confers robustness to noise and accents, yielding the best generalization and calibration; neutral remains challenging. The *CNN* (mel-spectrogram input) lacks temporal/context modelling, causing failures on subtle classes; its efficiency is attractive for edge devices but calibration is weak.

4.9 Deployment and Inference

The deployed pipeline ingests YouTube/Drive links, extracts audio, and infers emotions via all trained models; the final output is taken from the best performer (HuBERT) using confidence scores. Example predictions show a dominant class with ~0.515 confidence and a distribution indicating competing high-arousal states (anger/happiness) with low neutral/sad probabilities; temporal plots reveal fluctuation across a 250 s clip, capturing shifts in affect. An optional ASR+summarization path (Whisper + DistilBART) produces a transcript (≈2.9k chars) and a concise summary saved to file, extending SER into semantic understanding.

```
Aggregated Emotion (full audio): ang | Confidence: 0.5155 |
```





5. Discussion

5.1 Implications for Educational

Use SER can complement existing student-support workflows by flagging sessions indicative of stress, frustration, or prolonged low engagement. Calibrated probabilities enable cautious thresholding and human-in-the-loop validation. Summaries provide context while avoiding full replay of sessions, aiding counsellors and tutors.

5.2 Why HuBERT Generalised Best

HuBERT's masked prediction of discrete speech units encourages representations that are robust to noise and accents, which are common in user-generated classroom audio. This advantage translated into both higher macro-F1 and more reliable confidence estimates than the alternatives evaluated.

5.3 Ethical, Legal, and Social Considerations

Deployed SER systems must address privacy, consent, fairness, and transparency. In educational settings, explicit informed consent, opt-out options, data minimisation, and secure storage are essential. Models should be monitored for bias across accents, genders, and cultural groups. Outputs must be framed as decision-support—not diagnosis—and reviewed by trained staff. Governance should include DPIAs, retention limits, and clear incident-response procedures.

5.4 Limitations

The dataset size is modest and English-only; weak labels inherit biases from the initial classifier; and results are specific to four emotion categories. Colab-scale training constrained batch sizes and epochs. Whisper accuracy can degrade with heavy noise or strong accents. We did not evaluate costed real-time deployment or multimodal fusion.

5.5 Future Work

Future directions include:

(i) expanding multilingual corpora with institution-consented, high-quality audio;

(ii) probabilistic/soft labels and active learning to mitigate subjectivity;

(iii) knowledge distillation and quantisation for edge deployment;

(iv) richer dashboards for longitudinal monitoring; and

(v) multimodal fusion (text + audio, optional video) with privacy-preserving techniques.

6. Conclusion

We presented a practical SER pipeline tailored for student–teacher dialogues, compared leading transformer encoders and a compact CNN, and demonstrated that HuBERT offers the most balanced performance and calibration on naturalistic data. By integrating ASR and summarisation, the system converts raw audio into concise, reviewable insights to support student wellbeing. Responsible deployment demands strong governance, calibration-aware thresholds, and human oversight.